

RESEARCH ARTICLE



Adverse event reporting following immunization of hepatitis B vaccine: A 13-year review

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ABSTRACT

Hepatitis B vaccination is the most effective means of interrupting HBV transmission. Although the hepatitis B vaccine is very effective and safe, adverse events following immunization do occur and need to be reported so that problems can be identified and appropriate corrective action can be taken. Most of the research on AEFI focuses on the safety observation of newly used vaccines, and there are few long-term studies on AEFI of the hepatitis B vaccine. This study retrospectively analyzes the reporting rate, clinical symptoms, and onset time of AEFI of the hepatitis B vaccine in Quzhou from 2011 to 2023, and compares the differences in AEFI reporting rates between different types of hepatitis B vaccines, different vaccination ages, and different doses. The surveillance results show that from 2011 to 2023, the AEFI reporting rate of hepatitis B Vaccines in Quzhou was 17.55/100,000 doses. 98.73% of reported AEFI were non-serious. The types of AEFI reported were vaccine product-related reactions, immunization anxiety-related reactions, and coincidental events. 94.12% of vaccine product-related reactions occurred within 3 days, and the main symptoms were fever, local reactions at the injection site, and rash. The AEFI reporting rate of the CHO vaccine was higher than that of the yeast vaccines, and the probability of AEFI in children under 1 year of age receiving the hepatitis B vaccine was higher in the latter dose than in the previous dose. The 13-year-long AEFI surveillance provides reliable evidence of the safety of the hepatitis B vaccine.

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Introduction

Hepatitis B virus (HBV), which causes both acute and chronic liver disease, is a serious global public health threat that significantly increases the global burden of disease and mortality. The World Health Organization (WHO) estimated in 2019 that 296 million people worldwide had chronic HBV infection and more than 820,000 people died prematurely from liver cancer and cirrhosis.^{1,2} China is the country with the highest burden of HBV infection,³ with approximately 90 million people suffering from chronic HBV infection, and approximately 300,000 people dying each year from chronic HBV infection-related diseases.⁴ Hepatitis B vaccination is the most effective means of interrupting HBV transmission. In line with WHO recommendations, 190 countries around the world have included the hepatitis B vaccine in their national childhood immunization programs.⁵ Due to the widespread use of the hepatitis B vaccine, the global prevalence of chronic HBV infection in children under 5 years of age had declined from 4.7% pre-vaccination to less than 1% in 2019.⁶ Compared with the prevaccine era, chronic HBV infection in China had reduced by 90% among children <15 years of age and by 97% among children <5 years of age in 2014.⁷

More than 1 billion doses of hepatitis B vaccine have been used worldwide since 1982.⁶ A large number of clinical trials

and extensive practical applications have proven that the hepatitis B vaccine is very effective and safe. Although vaccines are among the safest medicines, adverse events following immunization do occur and need to be reported so that problems can be identified and appropriate corrective action can be taken.⁸ Lack of trust in vaccine safety and concerns about adverse reactions can cause parents to hesitate to vaccinate their children or even refuse to be vaccinated.⁹ Conducting safety surveillance can minimize the occurrence of adverse reactions and negative impacts on the public, reduce vaccine hesitation to ensure continued public confidence in vaccination, and ultimately ensure high immunization coverage, thereby reducing illness and death from vaccine-preventable diseases.

Adverse Events After Immunization (AEFI) surveillance is a key component of the vaccine safety surveillance system. An AEFI is any adverse event following immunization that is thought to be caused by the immunization. AEFI can be a true adverse event, i.e. a true consequence of immunization, or a coincidental event that is not due to immunization but is temporally related to the immunization. Occasionally, a series of coincidental adverse events may occur following vaccination that may threaten the acceptability of the immunization program. The most recent

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example is the adverse event associated with 17 infant deaths following hepatitis B vaccination in China in November 2013. An unconfirmed adverse event following hepatitis B vaccination, which resulted in the suspension of safe vaccination, greatly undermining the confidence of physicians and the public in the safety of the hepatitis B vaccine.^{10,11} Although the adverse event was eventually confirmed as a coincidence and allergy by a joint investigation by experts from the WHO and China's National AEFI Committee, the panic caused by it led to a 30% drop in the hepatitis B vaccine coverage rate in December 2013 compared with the same period in 2012, posing a threat of infection to the health of newborns.¹¹ Assuring the public that AEFI is under surveillance and actions are being taken to reduce the risk is critical to preventing vaccine panic and building trust. WHO recommends that all countries systematically collect, analyze, and evaluate AEFI to respond appropriately and quickly to reduce the negative impact on individual health and immunization programs. China first started conducting pilot AEFI surveillance in some provinces in 2005.¹² In June 2010, the Ministry of Health of China and the State Medical Products Administration jointly issued the Surveillance Program for Adverse Events Following Immunization in China, officially implementing AEFI surveillance nationwide. All AEFIs are reported to the surveillance system by healthcare providers, screened, and triaged by trained immunization program staff, who verify the diagnosis and timing of immunization and classify them according to standardized national case definitions. The year 2011 was the first year in which China officially launched national AEFI surveillance and carried out standardized surveys and causality assessments, and since that year China's AEFI surveillance indicators have all met or exceeded WHO's assessment standards.

Currently, most of the research on AEFI focuses on the safety observation of newly used vaccines,^{13–16} and there are few long-term studies on AEFI of the hepatitis B vaccine. Using the data from the formal implementation of AEFI surveillance, this study retrospectively analyzed the reporting rate, clinical symptoms, and onset time of AEFI following hepatitis B vaccination in Quzhou from 2011 to 2023, and compared the differences in AEFI reporting rates between different types of hepatitis B vaccines, different vaccination ages, and different doses, aiming to further provide evidence for the safety of hepatitis B vaccines.

Materials and methods

Data sources

The AEFI data for 2011–2023 were obtained from the Chinese AEFI surveillance system, and the vaccination data were obtained from the National Immunization Program Information Management System. In cases where individuals received two or more vaccines simultaneously, we attributed the reported AEFI to the reported suspected vaccine according to the following principles: (1) Local reactions at the injection site were identified through vaccination records; (2) For systemic reactions, reported AEFIs were

attributed to all vaccines administered concomitantly. Before the data were analyzed, the database was reviewed, duplicate records were eliminated, and inconsistencies were corrected.

AEFI classification

The AEFIs are divided into five categories: (1) vaccine product-related reactions, (2) vaccine quality defect-related reactions, (3) immunization error-related reactions, (4) immunization anxiety-related reactions, and (5) coincidental events. All AEFIs are assessed as a non-serious reaction or serious reaction: (1) non-serious reaction, a situation where no intervention is necessary or there is a physician visit or an event that interferes with daily activities or results in loss of working hours, (2) serious reaction, a situation with any untoward medical occurrence that results in death, hospitalization, prolongation of hospitalization, persistent or significant disability/incapacity, and life-threatening conditions or birth defects.

Statistical analysis

Case data exported from the AEFI surveillance system are Excel files. The reporting rate and composition ratio of AEFI were calculated by using Excel. The chi-square test was used in SPSS17.0 to compare the differences in reporting rates between different vaccine types, different vaccination ages, and different doses, a *P*-value less than 0.05 was considered statistically significant.

Results

From 2011 to 2023, a total of 157 cases of AEFIs of hepatitis B vaccine have been reported in the AEFI surveillance system, and 894,574 doses of hepatitis B vaccine were administered during the same period, with a reporting rate of 17.55/100,000 doses. The lowest AEFI reporting rate was in 2011 and the highest was in 2023. The reported AEFI categories were three: vaccine product-related reactions, immunization anxiety-related reactions, and coincidental events. There were no reports of vaccine quality defect-related reactions or immunization error-related reactions. Among them, the rate of reported vaccine product-related reactions was 17.10/100,000 doses, the rate of reported immunization anxiety-related reactions was 0.34/100,000 doses, and the rate of reported coincidental events was 0.11/100,000 doses. The reporting rate and classification of AEFIs of the hepatitis B vaccine in Quzhou are shown in Table 1.

Of all AEFIs, 98.73% were recorded as non-serious (155 cases), including 152 cases of vaccine product-related reactions and 3 cases of immunization anxiety-related reactions. The reporting rate of non-serious AEFI was 17.33/100,000 doses.

1.27% were recorded as serious AEFI (2 cases), and the reporting rate of serious AEFI was 0.22/100,000 doses. There was one death, with a mortality rate of 0.11/100,000 doses. The serious AEFIs recorded were 1 vaccine product-related reaction (thrombocytopenic purpura) and 1 coincidental event (death). The AEFI for this death was recorded as a 39-day-

Table 1. The reporting rate and classification of AEFI.

years	Vaccination doses	Number of AEFI reports	Reporting rate (/100,000 doses)	Classification				
				vaccine product-related reactions	vaccine quality defect-related reactions	immunization error-related reactions	immunization anxiety-related reactions	coincidental events
2011	72343	4	5.53	4	0	0	0	0
2012	83292	11	13.21	11	0	0	0	0
2013	75368	13	17.25	12	0	0	0	1
2014	68710	14	20.38	14	0	0	0	0
2015	71853	7	9.74	7	0	0	0	0
2016	82333	15	18.22	15	0	0	0	0
2017	90426	17	18.80	17	0	0	0	0
2018	74405	15	20.16	15	0	0	0	0
2019	67765	14	20.66	13	0	0	1	0
2020	56062	14	24.97	14	0	0	0	0
2021	48022	6	12.49	6	0	0	0	0
2022	48677	10	20.54	9	0	0	1	0
2023	55318	17	30.73	16	0	0	1	0
Total	894574	157	17.55	153	0	0	3	1

old infant who died 24 hours after hepatitis B and Bacillus Calmette Guérin co-administration. The autopsy showed that the main pathology was congenital heart disease and the main cause of death was heart failure.

Among vaccine product-related reactions, the highest reporting rate was fever at 9.95/100,000 doses, followed by redness and swelling at 4.36/100,000 doses, rash at 3.13/100,000 doses, and induration at 2.24/100,000 doses. The lowest reporting rate was thrombocytopenic purpura at 0.11/100,000 doses.

Most of the vaccine product-related reactions AEFIs occurred within a short time interval of vaccination, with 60.13% of AEFIs occurring within 24 hours and 94.12% within 3 days. In terms of vaccination form, 50% of the AEFI occurred within 24 hours with given hepatitis B vaccine single, 65% of the AEFI occurred within 24 hours with co-administration, 92% of the AEFI occurred within 3 days with given hepatitis B vaccine single, and 95.15% of the AEFI occurred within 3 days with co-administration. The time interval between vaccination and the onset of vaccine product-related reactions is shown in Table 2.

The differences in AEFI reporting rates among different groups were compared. The AEFI reporting rate of the CHO vaccine was higher than that of the yeast vaccine, and the difference was statistically significant. The AEFI reporting rate in people over 1 year old was higher than that in children under 1 year old, but the difference was not statistically

significant. There is a statistically significant difference in AEFI reporting rates between different vaccination doses in children under 1 year old. The AEFI reporting rate increases with the number of vaccination doses, with the lowest reporting rate for the first dose and the highest for the third dose. Comparison of AEFI reporting rates of hepatitis B vaccine product-related reactions among different vaccine types, vaccination ages, and vaccination doses are shown in Table 3.

Discussion

Conducting standardized AEFI investigations and causal relationship assessments and maintaining the surveillance capacity and sensitivity for AEFI of the hepatitis B vaccine will help enhance public confidence in hepatitis B immunization. The 13-year AEFI surveillance of the hepatitis B vaccine in Quzhou showed that adverse events were usually mild and transient, non-serious, such as fever, local injection site reactions, and rash. Records of serious reactions were extremely rare. The AEFI surveillance of the hepatitis B vaccine in Quzhou provides reliable evidence for the safety of the hepatitis B vaccine in China.

Data from the AEFI Surveillance System indicate that the reported rate of AEFIs of hepatitis B vaccine in Quzhou from 2011–2023 was 17.55/100,000 doses. The reported rate of AEFIs of hepatitis B vaccine in Guangdong Province was 3.54/100,000 from 2005 to 2017,¹⁷ and the reported rate of

Table 2. The time interval between vaccination and the onset of vaccine product-related reactions (classified by clinical symptoms and vaccination form).

Variable	<1day		1~day		4~day		7~day		Total	
	Number of reports	Composition ratio (%)	Number of reports	Composition ratio (%)	Number of reports	Composition ratio (%)	Number of reports	Composition ratio (%)	Number of reports	Composition ratio (%)
Clinical symptoms										
Fever	60	69.77	25	29.07	1	1.16	0	0	86	100
Redness and swelling	15	38.46	18	46.15	4	10.26	2	5.13	39	100
Rash	20	71.43	7	25	0	0	1	3.57	28	100
Induration	7	35.00	9	45.00	3	15.00	1	5.00	20	100
Thrombocytopenic purpura	0	0	0	0	0	0	1	100	1	100
Vaccination form										
single	25	50	21	42	3	6.00	1	2.00	50	100
co-administration	67	65.05	31	30.10	2	1.94	3	2.91	103	100
Total	92	60.13	52	33.99	5	3.27	4	2.61	153	100

Table 3. Comparison of AEFI reporting rates of hepatitis B vaccine product-related reactions among different vaccine types, vaccination ages, and vaccination doses.

Variable	Vaccination doses	Number of reports	Reporting rate (/100,000 doses)	Chi-square value	P-value
vaccine types				5.892	.015
yeast vaccine	846468	138	16.30		
CHO vaccine	48106	15	31.18		
vaccination ages				3.699	.054
<1 year	812039	132	16.26		
≥1 year	82535	21	25.44		
Doses for children <1 year				151.599	<.001
First dose	267526	1	0.37		
Second dose	276575	22	7.95		
Third dose	267938	109	40.68		

AEFIs of hepatitis B vaccine in China was 14.8/100,000 doses in 2020.¹⁸ The sensitivity of AEFI surveillance of the hepatitis B vaccine in Quzhou is higher than the national average. Compared with the reporting rate of AEFIs of hepatitis B vaccine of 11.8/100,000 doses in the United States from 1991 to 2001 and 10.4/100,000 doses in Australia from January 1, 2000, to September 30, 2002, the reporting rate of AEFIs of the hepatitis B vaccine in Quzhou was higher. As a result of the hepatitis B vaccine safety incident in November 2013, public sensitivity to hepatitis B vaccine safety increased, and AEFIs of hepatitis B vaccine reporting rate (20.38 per 100,000 doses) in 2014 saw a small spike. From 2020 to 2022, Quzhou experienced a three-year lockdown due to COVID-19. During this period, the dose of hepatitis B vaccination decreased, and the reporting rate of AEFIs of the hepatitis B vaccine fluctuated within a small range. In 2023, after the lockdown was lifted, the hepatitis B vaccination dose increased slightly, and AEFIs of the hepatitis B vaccine reporting rate (30.73/100,000 doses) reached the highest. Overall, there was an increasing trend in the reporting of AEFIs of hepatitis B vaccine over the 13 years. This may be related to the increasing public awareness of vaccine safety and the increased sensitivity of surveillance systems.¹⁷

Surveillance data over 13 years showed that the vast majority of AEFIs of hepatitis B vaccine were non-serious. Two cases of serious AEFI were reported, one of which was a relatively rare thrombocytopenic purpura. Thrombocytopenic purpura is usually caused by antibodies against the platelet glycoprotein IIb/IIIa complex¹⁹ and can be triggered by drugs, infections, cancer, and autoimmune diseases, with one of the rare triggers being vaccination.²⁰ This case of AEFI was found to have petechiae and ecchymosis 8 days after vaccination and recovered several days later. Another case of serious AEFI was a post-vaccination death, which was later confirmed to be an unfortunate coincidental event. The AEFI surveillance system in China for 2013–2020 reported 0.9 deaths per million doses,²¹ while an earlier study in the United States reported mortality rates ranging from 1.0 to 8.6 per million doses following hepatitis B vaccination.²² The AEFI mortality rate of hepatitis B vaccine in this paper is consistent with these reports.

The main symptoms of AEFIs for vaccine product-related reactions were fever, redness and swelling, rash, and induration. Fever is the most commonly reported vaccine product-related adverse reaction following hepatitis B vaccination. This is part of the body's immune response, a useful mechanism that promotes cell, blood vessel, and metabolic defense

mobilization, kills viruses, and activates immunity, but it turns into a “disease” above a certain temperature.²³ Because the aluminum adjuvants in the hepatitis B vaccines can increase local inflammatory responses and immune system responses,²⁴ redness swelling, and induration at the recipient's injection site are also common. But it may also have to do with the method of administration. The hepatitis B vaccine is not fully shaken before injection, and the large particles contained in the vaccine are slowly absorbed, leading to a local reaction. In addition, shorter needles or shallow injections increase the risk of vaccine-related local reactions. It is recommended to strengthen the training of vaccinators, standardize the operation of vaccination, and reduce the occurrence of local reactions. Rashes, including urticaria, measles scarlatina-like rash, and maculopapular, are also more commonly reported and are usually mild and do not require special treatment.

In terms of the time interval between vaccination and the occurrence of vaccine product-related reactions, the vast majority of reactions occurred within 3 days. This suggests that parents need to be reminded to observe children's reactions within 3 days after vaccination. The time of reaction following vaccine co-administration was faster, and the proportion of reactions occurring within 24 hours and 3 days was relatively higher than that of single. Because we could not distinguish between completed doses of hepatitis B single or co-administration, we were unable to compare the difference in AEFI reporting rates between the two. However, most studies found no statistically significant differences in the reporting rates of any AEFI following vaccine single and co-administration.²⁵ Vaccine co-administration is an effective vaccination strategy associated with high coverage and vaccine timeliness.²⁶ All recommended pediatric routine immunizations can be co-administered, and there is no recommendation against co-administration unless reported in the summary of product characteristics. Childhood vaccine co-administration is an effective strategy to maintain or even increase vaccination rates.²⁷

Three cases of immunization anxiety-related reactions were reported in 13 years. Anxiety-related AEFIs are related to “anxiety” about immunization. Individuals may experience fear- or anxiety-related reactions in anticipation of or as a result of any injection,²⁸ independent of the vaccine itself.

There was a statistically significant difference in the reporting rate of AEFI following different types of hepatitis B vaccination. Over the past 13 years, 94.32% of the hepatitis B vaccines used in Quzhou were yeast cell vaccines and 5.68% were Chinese hamster ovary cell (CHO) vaccines.

The yeast vaccine is a vaccine produced using yeast cells, while The CHO vaccine is a vaccine produced using hamster ovary cells. Both of them are recombinant genetically engineered vaccines, the difference mainly lies in the difference in the recombinant cell material. The reporting rate of the CHO vaccine product-related reactions was higher than that of the yeast vaccines, and the difference was statistically significant. This may be related to the higher specifications and dosage of CHO vaccines. The commonly used clinical specifications for yeast vaccines are generally 10 µg/0.5 mL/vial, while the commonly used clinical specifications for CHO cells are generally 20 µg/1 mL/vial. In an open-label randomized trial in China, the AEFI reporting rate was comparable between 20 µg of yeast hepatitis B vaccine and 20 µg CHO hepatitis B vaccine.²⁹

The reporting rates of vaccine product-related reactions following hepatitis B vaccination vary among people of different ages. Reported rates of vaccine product-related reactions following hepatitis B vaccination were slightly higher in people aged 1 year and above than in children younger than 1 year of age, but the difference was not statistically significant. There is a high probability that people aged 1 year and above are booster-vaccinated or are in the older age group that did not receive the hepatitis B vaccine in infancy. Recipients of booster doses of hepatitis B vaccine receive repeated stimulation at the injection site, resulting in increased sensitivity to vaccine antigens or adjuvants. And older people who get vaccinated tend to be more responsive to the vaccine and more observant.

The reporting rates of vaccine product-related reactions vary with different doses of hepatitis B vaccine. The probability of AEFI was significantly higher for the later dose of the 3-dose hepatitis B vaccine than for the previous dose in children younger than 1 year of age. Subsequent doses may be more likely to elicit a response due to sensitization of the body at earlier doses.³⁰ Studies of adverse events following COVID-19 vaccination have also reported that systemic adverse reactions are usually reported more frequently after the second dose than after the first dose.³¹

The standardized surveillance of AEFI in Quzhou has spanned 13 years, and we searched all reports of AEFIs following hepatitis B vaccination during this period, allowing for a comprehensive assessment. High-quality and sensitive AEFI records improve the reliability of surveillance analysis and provide a tool for evaluating the safety of the hepatitis B vaccine. Our description and analysis of AEFI further confirm the safety of the hepatitis B vaccine. This is important for increasing public confidence in hepatitis B vaccination and promoting adherence to the vaccination program. We found that more than 90% of AEFI occurred within three days following vaccination, especially for co-administration, the proportion of AEFI occurring within three days was even higher; the AEFI reporting rate following the third dose was significantly higher than that of the first and second doses. The surveillance results are a good reminder that we need to do a good job of observation and emergency management based on these characteristics in practice. We also found that the 20 µg CHO hepatitis B vaccine had a higher AEFI reporting rate than the 10 µg yeast hepatitis B vaccine; people over 1 year old had a higher AEFI reporting rate of hepatitis B vaccine than those under 1 year old. Our study has guiding significance

for the selection of the hepatitis B vaccine in the future. When vaccinating against hepatitis B, it would be a better choice to complete the vaccination of 10 µg yeast hepatitis B vaccine within 1 year of age.

Active surveillance in China is still at an initial exploratory stage, with no effective active surveillance mechanism covering the whole country, and a further lack of application of advanced methods and technologies for active surveillance. Compared to the accurate and huge cost of active surveillance, passive surveillance provides extensive and economical coverage. Our data came from passive surveillance. Although it has limitations such as underreporting, late reporting, and difficulty in conducting causal analysis, it is still the most feasible method for AEFI surveillance in China.¹⁸ Another limitation of this study is that our data were collected from Quzhou and cannot represent the whole of China.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Xiaoying Gong, the chief physician, has been engaged in disease prevention and control for 28 years. Engaged in hepatitis B control work for more than 10 years, responsible for AEFI investigation and management after hepatitis B vaccination. She has rich experience in infectious disease prevention and control and has published many papers at home and abroad, including one SCI paper.

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Data availability statement

All data can be available upon request from the corresponding author.

Author contributions disclaimer

Xiaoying Gong wrote the article and Zhiying Yin critically reviewed the article; Quanjun Fang, Jianyue Zhong, and Canjie Zheng were involved in the investigation and data analysis. All authors have agreed on the journal to which the article will be submitted and all agree to take responsibility for all aspects of the work.

Institutional review board statement

The study was conducted in accordance with the Declaration of Helsinki and was approved by the Ethics Committee of the Quzhou Center for Disease Control and Prevention.

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